

Connecting to Shelton's LocalStack AWS Simulation via ngrok

Team setup guide for remote AWS CLI, SDK, Terraform and optional kubectl access

Security note: Shelton will share the actual ngrok URL separately and only with the intended demo participants. Do not post the URL in public chats, repositories, screenshots, or recordings.

1. What this guide is for

Shelton will run LocalStack Pro on his macOS machine. LocalStack simulates AWS services locally, and ngrok exposes selected local ports through temporary public endpoints so remote team members can connect during the demo.

For most demo activities, you will connect your AWS tooling to a single ngrok HTTPS endpoint that forwards to LocalStack's AWS edge port. If direct Kubernetes access is needed, Shelton may also share a separate ngrok TCP endpoint for kubectl.

2. What Shelton will provide separately

- The AWS API ngrok HTTPS endpoint, for example: `https://REDACTED.ngrok-free.app`
- Optional: a username/password if ngrok basic authentication is enabled.
- Optional: a Kubernetes API TCP endpoint for kubectl, for example: `0.tcp.ngrok.io:12345`
- Optional: a kubeconfig file or kubeconfig snippet if kubectl access is part of the demo.
- The LocalStack region to use, usually `us-east-1` unless stated otherwise.

3. What you need installed

Tool	Required?	Purpose
AWS CLI v2	Yes	Run AWS commands against the LocalStack endpoint.
kubectl	Only if requested	Connect to the simulated EKS/Kubernetes cluster.
Terraform	Only if requested	Run infrastructure examples against LocalStack.
curl	Recommended	Quick endpoint connectivity test.

4. Configure your shell for the demo

Replace the placeholder value below with the actual ngrok URL that Shelton shares privately.

```
export LOCALSTACK_ENDPOINT="https://REDACTED.ngrok-free.app"
export AWS_ACCESS_KEY_ID="test"
export AWS_SECRET_ACCESS_KEY="test"
export AWS_DEFAULT_REGION="us-east-1" "af-south-1"
```

If ngrok basic authentication is enabled

Your browser or tool may prompt for credentials. If using curl, include the username and password that Shelton provides:

```
curl -u demo:REDACTED "$LOCALSTACK_ENDPOINT/_localstack/health"
```

5. Connectivity checks

First, check that the public endpoint is reachable:

```
curl "$LOCALSTACK_ENDPOINT/_localstack/health"
```

Then check that AWS CLI can talk to LocalStack:

```
aws --endpoint-url="$LOCALSTACK_ENDPOINT" sts get-caller-identity
```

Expected style of result:

```
{
  "UserId": "test",
  "Account": "000000000000",
  "Arn": "arn:aws:iam::000000000000:user/test"
}
```

6. Useful AWS CLI commands for the demo

Use the same endpoint URL for each AWS CLI command:

```
aws --endpoint-url="$LOCALSTACK_ENDPOINT" eks list-clusters
aws --endpoint-url="$LOCALSTACK_ENDPOINT" eks describe-cluster --name demo-cluster
aws --endpoint-url="$LOCALSTACK_ENDPOINT" ec2 describe-vpcs
aws --endpoint-url="$LOCALSTACK_ENDPOINT" ecr describe-repositories
aws --endpoint-url="$LOCALSTACK_ENDPOINT" iam list-roles
```

7. Optional Terraform provider configuration

If the demo includes Terraform, configure the AWS provider to use Shelton's ngrok endpoint. Keep credentials as test values; do not use real AWS credentials for this LocalStack demo.

```
provider "aws" {
  region = "us-east-1"
  access_key = "test"
  secret_key = "test"
  skip_credentials_validation = true
  skip_metadata_api_check = true
  skip_requesting_account_id = true

  endpoints {
    ec2 = "https://REDACTED.ngrok-free.app"
    eks = "https://REDACTED.ngrok-free.app"
    ecr = "https://REDACTED.ngrok-free.app"
    iam = "https://REDACTED.ngrok-free.app"
    sts = "https://REDACTED.ngrok-free.app"
  }
}
```

8. Optional kubectl access

Only complete this section if Shelton says direct Kubernetes access is part of the demo. AWS API access and kubectl access are different: AWS CLI talks to the LocalStack AWS edge endpoint, while kubectl talks directly to the simulated Kubernetes API server.

Shelton may share a separate TCP endpoint such as 0.tcp.ngrok.io:12345 and a kubeconfig. If you are asked to create the kubeconfig manually, use this template and replace the server value:

```
apiVersion: v1
kind: Config
clusters:
- cluster:
  server: https://0.tcp.ngrok.io:12345
  insecure-skip-tls-verify: true
  name: localstack-eks-demo
contexts:
- context:
  cluster: localstack-eks-demo
  user: localstack-demo-user
  name: localstack-eks-demo
current-context: localstack-eks-demo
users:
- name: localstack-demo-user
  user:
  token: demo
```

Save it as localstack-demo-kubeconfig.yaml, then test:

```
kubectl --kubeconfig ./localstack-demo-kubeconfig.yaml get nodes
kubectl --kubeconfig ./localstack-demo-kubeconfig.yaml get pods -A
```

Demo-only note: insecure-skip-tls-verify is used here because the LocalStack Kubernetes API certificate may not match the ngrok TCP hostname. Do not copy this pattern into production environments.

9. Troubleshooting quick checks

Symptom	What to check
Could not connect to endpoint URL	Confirm you pasted the latest ngrok URL. ngrok URLs can change unless a static/dev domain is used.
Access denied / auth prompt	Ask Shelton whether ngrok basic authentication is enabled and confirm the username/password.
AWS CLI tries real AWS	Make sure every command includes <code>--endpoint-url="\$LOCALSTACK_ENDPOINT"</code> or your SDK/provider's endpoint.
kubectl cannot connect	Confirm you are using the separate ngrok TCP endpoint, not the HTTPS AWS API endpoint.
Unexpected AWS account or region	Use <code>AWS_ACCESS_KEY_ID=test</code> , <code>AWS_SECRET_ACCESS_KEY=test</code> , and <code>AWS_DEFAULT_REGION=us-east-1</code> .

10. Safety rules for the demo

- Do not share the ngrok URL outside the demo group.
- Do not commit the ngrok URL, kubeconfig, or credentials to Git.
- Use dummy LocalStack credentials only: test/test unless Shelton says otherwise.
- Assume anything sent to the endpoint reaches Shelton's local demo environment.
- Close terminals and remove exported variables after the demo if they are no longer needed.

Reference notes

LocalStack uses a primary AWS edge endpoint, commonly port 4566, and can allocate external service ports in the 4510–4559 range for services that require direct external access. ngrok can expose local HTTP/S and TCP services through public endpoints. This guide intentionally redacts the real ngrok URL; Shelton will share it separately for safety.

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